



HR Attrition Analysis

Understanding Why Employees Are Leaving — TalentCore Solutions

Analyst

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Tools

Python · MySQL · Power BI

Dataset

IBM HR Analytics — Kaggle

Date

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The Problem We Are Solving

TalentCore Solutions is losing approximately **16% of its employees every year** — significantly above the healthy industry benchmark of 10%. The HR department lacks visibility into *who* is leaving, *why* they leave, and *when* they tend to leave.

16%

Attrition Rate

vs. 10% industry benchmark

237

Employees Lost

in the last year alone

₹1.18Cr

Annual Cost

₹50,000 replacement cost per employee



No data-driven retention strategy currently exists at TalentCore Solutions.

About The Data

Dataset Details

Source: IBM HR Analytics Employee Attrition (Kaggle)

- 1,470 employee records
- 32 features per record
- Zero null values — clean dataset
- Target variable: Attrition {Yes/No}

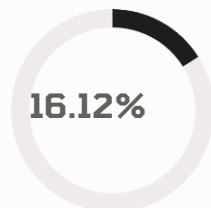
Key Column Categories

- **Employee Info:** Age, Gender, Education, Marital Status
- **Job Info:** Department, Job Role, Job Level, Years at Company
- **Satisfaction:** Work-Life Balance, Environment Satisfaction
- **Compensation:** Monthly Income, Stock Option Level
- **Workload:** Overtime, Business Travel

Tools Used

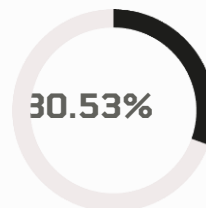
Python (Pandas, Matplotlib, Seaborn) · MySQL · Power BI

Exploratory Data Analysis — Key Findings



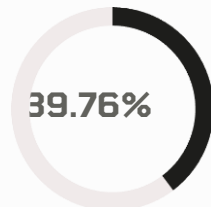
Overall Attrition

237 of 1,470 employees left — 6% above the 10% benchmark



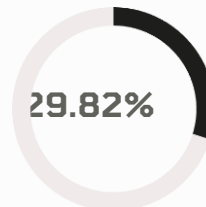
Overtime Attrition

Employees working overtime leave at 3x the rate



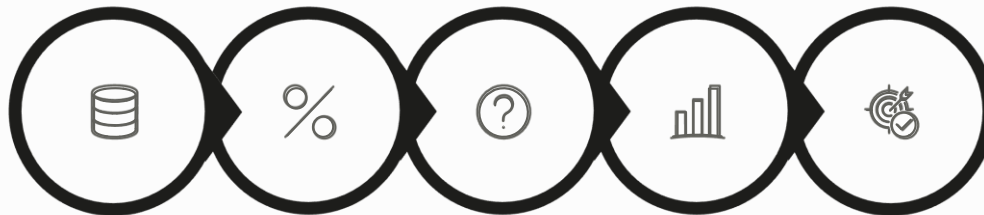
Sales Rep Attrition

Highest attrition of any job role in the company



New Joiner Risk

Employees with 0-2 years tenure are most at risk



Load Data

Attrition Rate

16 Questions

16 Visuals

Top 5 Drivers



Who Is Leaving TalentCore?

By Job Role

- **Sales Representative: 39.76%** — Highest
- Laboratory Technician: 23.94%
- Human Resources: 23.08%
- Sales Executive: 17.48%
- Research Scientist: 16.10%

By Department

- **Sales: 20.63%** — Highest
- Human Resources: 19.05%
- R&D: 13.84% — Lowest

By Age Group

- **Age 18–25: 34.78%** — Highest
- Age 26–35: 19.14%
- Age 46–60: 12.45%
- Age 36–45: 9.19% — Lowest

By Education Field

- **Human Resources: 25.93%** — Highest
- Technical Degree: 24.24%
- Marketing: 22.01%
- Medical: 13.58% — Lowest



Sales Representatives and young employees (18–25) are the most at-risk groups in the company.

Why Are Employees Leaving?

Overtime — 3× Risk

Overtime YES: **30.53%** attrition vs. Overtime NO: 10.44%. Employees working overtime are **3× more likely to leave**.

Work-Life Balance

Bad WLB: **31.25%** attrition — highest.
Good WLB: 16.86%. Better WLB: 14.22% — lowest.

Salary Gap — ₹2,045

Employees who **left** earned ₹4,787/month avg vs. those who **stayed** at ₹6,832/month — a ₹2,045 pay disparity.

Stock Options

No stock options: **24.41%** attrition.
Medium stock: just 7.59%. Stock options are a strong retention tool.

Environment Satisfaction

Low satisfaction: **25.35%** attrition. Very high satisfaction: 13.45%. Poor office culture doubles attrition risk.

When Are Employees Leaving?

By Tenure at Company

- **0–2 Years: 29.82%** — Critical risk period
- 3–5 Years: 13.82%
- 6–10 Years: 12.28%
- 11–20 Years: 6.67% — Lowest
- 20+ Years: 12.12%

By Years With Current Manager

- **0–1 Year: 28.32%** — Highest
- 2–3 Years: 14.20%
- 4–7 Years: 13.37%
- 8+ Years: 8.12% — Lowest

By Business Travel

- **Travel Frequently: 24.91%** — Highest
- Travel Rarely: 14.96%
- Non-Travel: 8.00% — Lowest

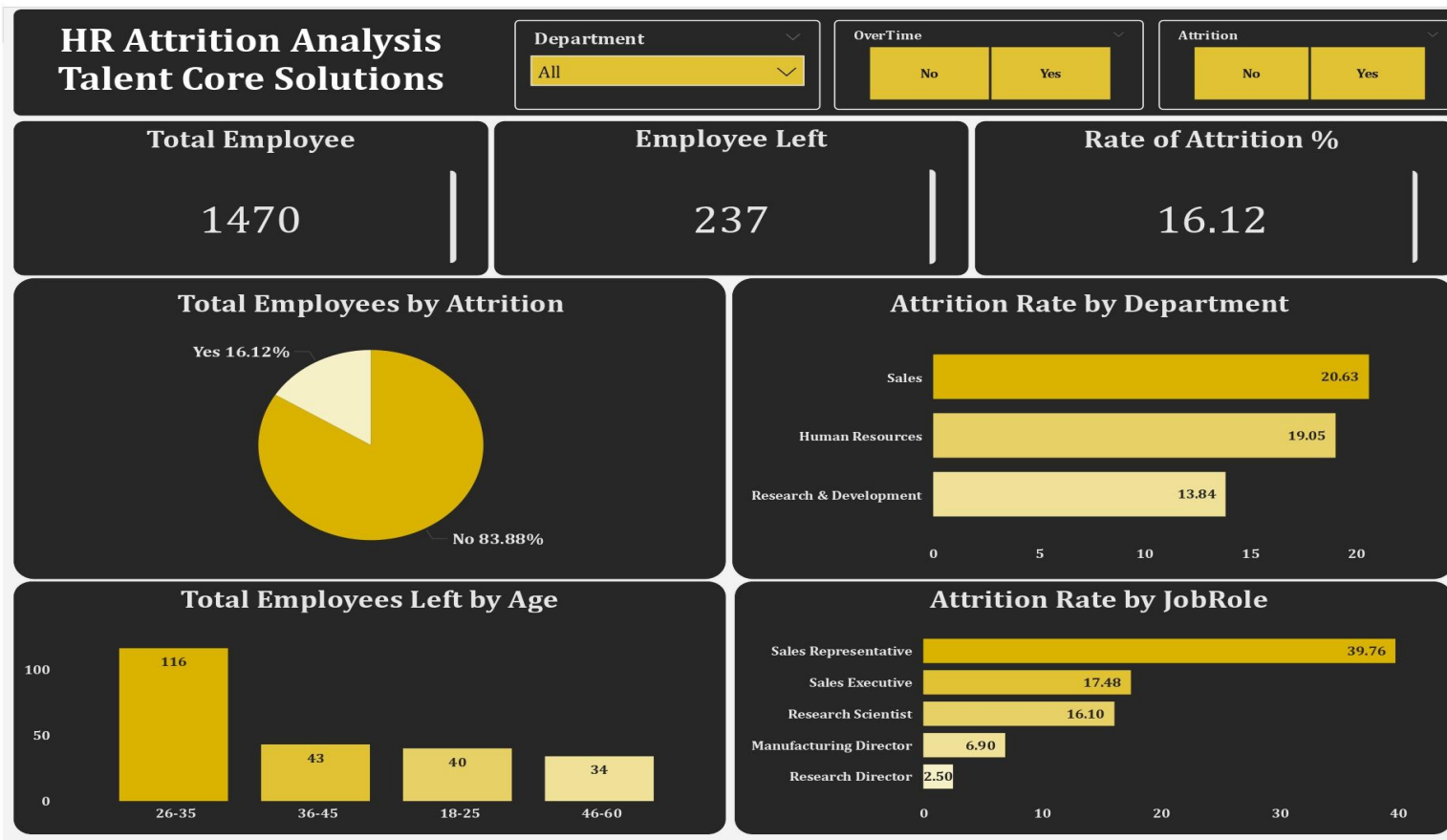
By Companies Worked

- **7+ Companies: 20.00%** — Highest (job hoppers)
- 2–3 Companies: 10.49% — Lowest



The first 2 years at the company and the first year with a new manager are the two most critical risk periods for attrition.

Interactive Power BI Dashboard – OVERVIEW (WHO?)



Interactive Power BI Dashboard — WHY?



Interactive Power BI Dashboard – When?



What Should TalentCore Do?

1 0–3 Months

Overtime Policy: Cap at 10 hrs/week, introduce compensation. **Salary**

Review: Close ₹2,045 gap for Sales Reps; quarterly performance bonuses.

2 3–6 Months

Onboarding: Redesign first 2-year experience; assign mentors; monthly check-ins. **Manager Training:** Retention skills, structured handovers, mandatory 1-on-1s. **Stock Options:** Expand program to non-stock employees.

3 6–12 Months

Work-Life Balance: Flexible hours, strict disconnect policy, mental health programs. **Sales Focus:** Special retention program, reduce travel, quarterly incentives. **Promotion Policy:** Max 2–3 year cycle; transparent growth paths.



Impact of This Analysis

Current Situation

- Attrition Rate: **16.12%** (above 10% benchmark)
- Employees Lost: 237 per year
- Annual Replacement Cost: **₹1,18,50,000**
- No data-driven retention strategy

After Recommendations

- Target Attrition Rate: **Under 10%**
- Employees Saved: ~90 per year
- Projected Annual Savings: **₹45,00,000**
- Data-driven HR decisions across all departments

Key Metrics to Track

- Monthly attrition rate by department
- Overtime hours per employee
- Employee satisfaction survey scores
- New joiner retention at 6 months and 1 year

"By addressing overtime, salary gaps and onboarding experience — TalentCore Solutions can reduce attrition from 16.12% to under 10%, saving approximately ₹45 lakhs annually in replacement costs."